

Gravity, Matter, and Abelian Gauge Terminals

from One Ordered Tau Descent

Tau Core Foundation Paper VII: First Proof Packet

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August 2026

Abstract

We ask whether the already constructed Tau gravity, temporal, quantum, matter-occupation and Abelian gauge terminals can belong to one physical descent rather than to independently fitted sector models. The inherited results do not yet prove that Nature selects such a descent. They do, however, permit a finite conditional assembly. We define one ordered source ledger: the morphological body is solved and frozen first; geometry, occupied gauge and matter sectors are then varied on that body; observer restriction is applied last. A disjoint occupation ledger assigns every physical mode to exactly one classical or quantum stress export. Under diffeomorphism and $U(1)$ gauge invariance, stationarity gives the semiclassical Einstein–Maxwell–matter equations and their Ward identities. Their weak field limit is the Poisson equation sourced by the same once-counted total energy density, together with the ordinary Maxwell equations. We prove that a simultaneous body/post-body Hessian cannot retain nonzero body conditioning and exact no-backreaction, and that duplicate classical/quantum occupation necessarily double counts stress. Conversely, recovering standard terminal equations does not identify common Tau provenance: an equal-terminal independent-source counterfamily remains. The result is therefore a conditional source-assembly and non-double-counting theorem, not a derivation of quantum gravity, the Standard Model, physical constants or Nature-level source selection.

Inputs from earlier papers. Paper I supplies the body-first descent; Paper II the typed source support; Paper III recoverable record transport; Paper IV the parent-selection boundary; Paper V the observer clock and causal leaf; Paper VI the conditional quantum terminal. The theory hub supplies the P1 gravity–time–radiation ordered ledger and the P3 Abelian handoff.

Results owned here. The joint terminal ledger, the disjoint occupation rule, the Ward/Bianchi consistency theorem, the simultaneous-action and duplicate-stress no-go results, and the equal-terminal provenance no-go.

1 Question and claim boundary

Let M_τ be the stabilized atemporal morphological body. Paper VII asks:

$$\boxed{\begin{array}{l} \text{Can gravity, matter, gauge/radiation and quantum backreaction} \\ \text{descend from one once-counted ordered physical source ledger?} \end{array}} \quad (1)$$

The word *one* denotes common provenance and calibration. It does not mean one emitter, one scalar varied at every generative stage, or one terminal observable. The generative order remains

$$B_\tau^{\text{phys}} + s_U \longrightarrow M_\tau \longrightarrow \mathfrak{L}_\tau \longrightarrow \text{Desc}_O \longrightarrow \{g, A, \Phi, \rho_O\}. \quad (2)$$

No terminal residual may select M_τ or the ledger.

2 Inherited-results ledger

Result	Owner	Status and assumptions	Use here
Body-first descent	Paper I	conditional physical MVP	generative order
Typed incidence	Paper II	finite source theorem	occupied domains
Record recovery	Paper III	exact/approximate conditional theorem	observer terminal
Parent selection	Paper IV	present reduct does not select full packet	claim boundary
Temporal descent	Paper V	conditional clock/causal leaf	time and weak limit
Quantum terminal	Paper VI	conditional complex/effect/QFT packet	Γ_q and ρ_O
P1 assembly	theory hub	conditional metric/time/radiation ledger	g , null radiation, stress
P3 Abel handoff	theory hub	conditional $Z_F = p_* K_Y$	Maxwell/current sector

Table 1: Dependency and ownership ledger. No inherited conditional result is promoted to a Nature theorem.

3 The ordered Paper-VII ledger

Let y denote body coordinates, and write $z = (g, A, \Phi, \rho)$ for post-body geometry, Abelian gauge, occupied classical matter and quantum-state data.

Definition 1 (Once-counted ordered ledger). *The Paper-VII ledger is*

$$\mathfrak{L}_7 = (\mathcal{A}_{\text{pre}}[y; B, s], y_*, \Gamma_7[g, A, \Phi, \rho \mid y_*], \text{Desc}_{O \leftarrow y_*}, \mathcal{U}), \quad (3)$$

where $D_y \mathcal{A}_{\text{pre}}(y_*) = 0$, the physical body Hessian is positive, and

$$\Gamma_7 = \Gamma_{\text{grav}}[g \mid y_*] + \Gamma_{\text{EM}}[A, g \mid y_*] + \Gamma_{\text{occ}}[\Phi, A, g \mid y_*] + \Gamma_q[\rho, A, g \mid y_*] + \Gamma_{\text{ct}}[A, g]. \quad (4)$$

Each occupied mode and each counterterm appears in exactly one summand. Observer restriction is evaluated after stationarity and is not varied back against y_* . The unit packet $\mathcal{U}$ is shared by every summand.

The ordering is structural. If $H_B, H_P > 0$ and the post-body equations have body derivative C , the linear response is lower triangular:

$$\begin{pmatrix} \delta y \\ \delta z \end{pmatrix} = \begin{pmatrix} H_B^{-1} & 0 \\ -H_P^{-1} C H_B^{-1} & H_P^{-1} \end{pmatrix} \begin{pmatrix} \delta j_B \\ \delta j_P \end{pmatrix}. \quad (5)$$

Thus morphology conditions the physical terminals, while a later terminal load does not rebuild the body.

Proposition 1 (Simultaneous-action no-go). *For an invertible positive simultaneous Hessian*

$$\mathbb{H} = \begin{pmatrix} H_B & C^\dagger \\ C & H_P \end{pmatrix}, \quad (6)$$

exact vanishing of body response to every pure post-body load implies $C = 0$. Hence nonzero body conditioning and exact body freeze cannot both be encoded by one simultaneous unconstrained Hessian solve.

Proof. Schur inversion gives $\delta y = -H_B^{-1}C^\dagger(H_P - CH_B^{-1}C^\dagger)^{-1}\delta j_P$. If this vanishes for every δj_P , invertibility forces $C = 0$. $\square$

4 Disjoint occupation and stress ownership

Let the occupied mode labels form a disjoint union

$$\mathcal{I}_{\text{occ}} = \mathcal{I}_{\text{cl}} \sqcup \mathcal{I}_q. \quad (7)$$

Classical coherent modes represented in Γ_{occ} are excluded from the state-dependent part of Γ_q ; vacuum polarization and local renormalization terms occur only in $\Gamma_q + \Gamma_{\text{ct}}$. This disjoint partition is a sufficient transparent ledger, not the only possible field-theoretic organization. A background-field, one-particle-irreducible or two-particle-irreducible construction is also admissible when its mean field, fluctuation determinant and counterterms are derived from one effective action and no full contribution is inserted twice.

Proposition 2 (Duplicate-stress no-go). *If one nonzero occupied mode i is included in both $\mathcal{I}_{\text{cl}}$ and $\mathcal{I}_q$ with the same metric variation, its stress contribution is counted twice. No terminal redefinition removes this without changing the metric equation or adding a compensating subtraction.*

Proof. Metric variation is linear. The duplicated summands give $-2(\sqrt{-g})^{-1}\delta(\Gamma_i + \Gamma_i)/\delta g^{\mu\nu} = 2T_{\mu\nu}^{(i)}$. A subtraction merely moves the ownership entry and must therefore be recorded explicitly in Γ_{ct} . $\square$

5 Joint terminal theorem

Assume on the selected Lorentzian leaf

$$\Gamma_{\text{grav}} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} (R - 2\Lambda), \quad (8)$$

$$\Gamma_{\text{EM}} = -\frac{Z_F}{4} \int d^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu}, \quad Z_F = p_* K_Y, \quad (9)$$

and let the matter and renormalized quantum functionals be diffeomorphism and $U(1)$ gauge invariant. These are standard-limit forms inherited conditionally; Paper VII does not derive their Nature-level selection.

Theorem 1 (Common-descent Einstein–Maxwell–matter assembly). *Under the ordered ledger, disjoint occupation rule, common unit packet and the stated invariances, stationarity of Eq. (4) gives*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \left(T_{\mu\nu}^{\text{EM}} + T_{\mu\nu}^{\text{cl}} + \langle T_{\mu\nu}^q \rangle_{\text{ren}} \right), \quad (10)$$

$$\nabla_\mu (Z_F F^{\mu\nu}) = J_{\text{cl}}^\nu + \langle J_q^\nu \rangle_{\text{ren}}. \quad (11)$$

On the matter and gauge equations, the Ward identities and Bianchi identity imply

$$\nabla^\mu T_{\mu\nu}^{\text{tot}} = 0, \quad \nabla_\mu J_{\text{tot}}^\mu = 0. \quad (12)$$

No additional gravity, gauge or quantum-backreaction gain is introduced at the terminal stage.

Proof. Equations (10) and (11) are the metric and gauge Euler equations of the single post-body functional. Gauge invariance gives current conservation. Diffeomorphism invariance gives the combined stress Ward identity, including the Lorentz-force exchange between matter and gauge sectors. The divergence of the left side of Eq. (10) vanishes identically, so the equations are mutually consistent. Disjoint occupation and the unique counterterm ledger ensure that each contribution is varied once. $\square$

Corollary 1 (Standard weak-field leaf). *For $g_{00} = -(1 + 2\Phi_N/c^2)$, stationary weak fields and nonrelativistic sources, Eq. (10) reduces to*

$$\nabla^2 \Phi_N = 4\pi G \rho_{\text{tot}}, \quad (13)$$

where ρ_{tot} is the same once-counted density appearing in the ledger. Equation (11) reduces to the ordinary Maxwell equations after the inherited field normalization. This is standard recovery, not a Tau-specific deviation.

6 Local constraint structure

The covariant equations must reproduce constraints, not only evolution equations. Let n^μ be the unit normal to an observer-compatible spacelike leaf, $h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$ its induced metric, and K_{ij} its extrinsic curvature. With $\rho = n^\mu n^\nu T_{\mu\nu}^{\text{tot}}$ and $j_i = -h_i^\mu n^\nu T_{\mu\nu}^{\text{tot}}$, the normal projections of Eq. (10) give, in the stated sign convention,

$${}^{(3)}R + K^2 - K_{ij}K^{ij} = 2\kappa\rho + 2\Lambda, \quad (14)$$

$$D_j(K^{ij} - h^{ij}K) = \kappa j^i. \quad (15)$$

The normal projection of Eq. (11) gives the Abelian Gauss constraint

$$D_i(Z_F E^i) = \rho_J. \quad (16)$$

The Ward identities propagate these constraints when the classical and quantum matter equations hold. Paper VII therefore imports the standard local constraint architecture from the one joint functional; it does not derive a new constraint algebra or quantize it.

7 Primitive terminal inventory

The assembled packet contains several typed outputs, not several names for the same information:

Family	Primitive terminal content	Derived or composite descendants
Geometry/gravity	g , connection, curvature and constraints	proper time, tidal response, lensing after a null source
Gauge/radiation	A , F , conserved current, two screen polarizations	photon propagation, redshift and optical transport
Matter occupation	occupied localized or propagating solutions Φ	mass, energy, stress and charge records
Quantum operation	ρ , effects, instruments and correctable composition	Born statistics and renormalized quantum stress/current
Observer access	causal orientation, clock and stable restriction	observer-relative terminal classes

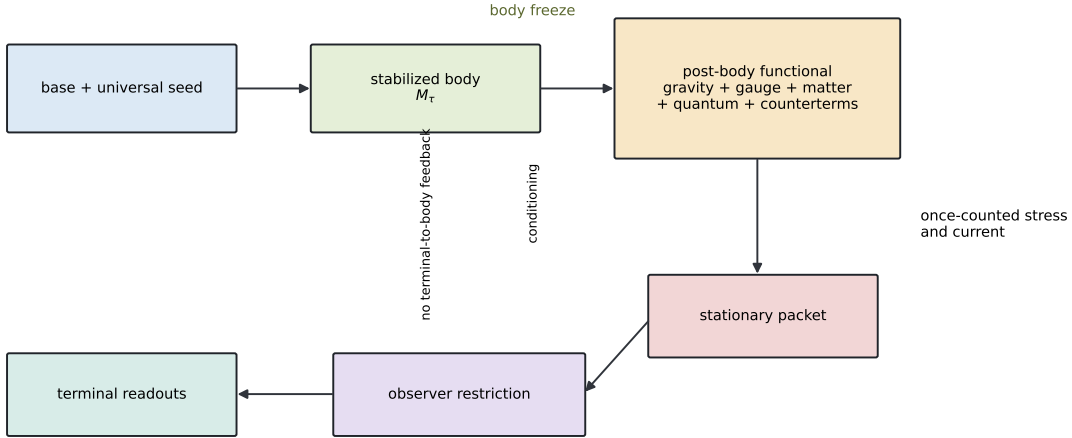


Figure 1: Ordered common-source construction. The body is stabilized before the post-body functional is solved; observer restriction is terminal.

Stress-energy, current, redshift and lensing are variational or composite outputs of these families; they are not silently counted as new primitive source directions. Conversely, the symmetric local gravity descriptor cannot stand in for internal gauge charge, quantum phase or occupied matter. A global information-completeness claim must wait for the non-Abelian and fermionic known-terminal completions.

8 Why the standard limit does not prove common provenance

Proposition 3 (Equal-terminal provenance no-go). *The equations (10)–(12) do not imply that their coefficients or sectors arise from one Tau source.*

Proof. For any nonzero constants a_G, a_F, a_M, a_q , independently rescale the four sector functionals and compensate by redefining their terminal couplings so that the displayed equations are unchanged. The terminal solution set is the same while the source ownership graph and common-unit identification differ. Therefore standard-limit agreement cannot be inverted into common provenance. $\square$

This no-go fixes the empirical interpretation: terminal data may test a source-frozen forward ledger, but may not choose its source partition, susceptibilities or unit map.

9 Quantum backreaction and the quantum-gravity boundary

The quantum term in Eq. (10) is a renormalized expectation value on a classical body-derived metric, in the standard semiclassical sense [Birrell and Davies, 1982, Wald, 1977]. Local covariance and admissible renormalization follow the usual curved-spacetime QFT framework [Hollands and Wald, 2001, Brunetti et al., 2003]. The metric can also be driven by a stress-noise second cumulant in a stochastic extension, but this does not quantize geometry.

Paper VII therefore establishes a conditional quantum–gravity *interface*, not quantum gravity. A path integral or operator algebra for the metric, gravitational gauge fixing, a physical inner product and a nonperturbative ultraviolet completion remain absent.

10 Finite decision ledger

Question	Verdict	Status
One simultaneous body/post-body Hessian supplies conditioning and freeze	Impossible unless the cross derivative vanishes	Exact no-go
One ordered ledger supplies both	Yes, Eq. (5)	Conditional theorem
Classical and quantum occupation may overlap silently	No; stress doubles	Exact no-go
Einstein, Maxwell and Ward equations share one post-body variation	Yes under the declared packet	Conditional theorem
Newton/Maxwell limits follow	Yes on the standard weak leaf	Conditional recovery
Standard-limit agreement proves Tau provenance	No	Exact no-go
Geometry is quantized	Not derived	Open
Hamilton, momentum and Gauss constraints follow	Yes, on the selected standard leaf	Conditional recovery
Non-Abelian gauge and Standard Model spectrum follow	Not derived	Open
Physical base–seed source selects $\mathfrak{L}_7$	Not proved	Primary blocker

11 Conclusion

Paper VII closes a specific assembly problem. The inherited primitive terminals can be placed in one ordered, once-counted functional without rebuilding the morphology from downstream loads. Diffeomorphism and Abelian gauge invariance then make gravity, current and stress conservation one consistent variational packet, while the disjoint occupation rule prevents a classical/quantum duplicate source. The Newtonian and Maxwell limits are recovered from that same packet.

The result remains conditional. Present Tau source laws do not yet prove that Nature occupies the ledger, fix its absolute constants, quantize the metric or select the non-Abelian and particle-spectrum completion. Those are the next finite source questions; they cannot be repaired by fitting composite lensing, galaxy or cosmological residuals.

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